THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 13: **Algebra Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The expression for '5 more than 2 times a number x' is: 1

(a) $5x + 2$ **(b)** $2x + 5$ **(c)** $2(x + 5)$ **(d)** $2 + 7x$

2. The value of $3x + 4$ when $x = 5$ is: 1

(a) 19 **(b)** 17 **(c)** 21 **(d)** 38

3. On simplifying $2x + 3 + 4x + 5$ we get: 1

(a) $14x$ **(b)** $6x + 8$ **(c)** $8x + 8$ **(d)** $6x + 15$

4. The two-digit number with tens digit x and ones digit 2 is written as: 1

(a) $2x$ **(b)** $20 + x$ **(c)** $x + 2$ **(d)** $10x + 2$

5. In a calendar, the four dates 3, 4, 10 and 11 form a 2×2 block. Their sum is: 1

(a) 20 **(b)** 30 **(c)** 28 **(d)** 26

6. In the expression $5x + 7$, the coefficient of x is: 1

(a) 12 **(b)** x **(c)** 5 **(d)** 7

7. The expression for '4 more than 3 times a number x' is: 1

(a) $3x + 4$ **(b)** $3(x + 4)$ **(c)** $4x + 3$ **(d)** $3 + 7x$

8. The value of $2x + 7$ when $x = 6$ is: 1

(a) 21 **(b)** 19 **(c)** 17 **(d)** 38

9. On simplifying $3x + 7 + 5x + 1$ we get: 1

(a) $16x$ **(b)** $8x + 8$ **(c)** $15x + 8$ **(d)** $8x + 7$

10. The two-digit number with tens digit x and ones digit 3 is written as: 1

(a) $3x$ **(b)** $30 + x$ **(c)** $x + 3$ **(d)** $10x + 3$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write an algebraic expression for: the sum of 7 and 5 times a number n. 2

12. Find the value of $3x + 2y$ when $x = 4$ and $y = 5$. 2

13. Simplify $2x + 3y + 5x + 4y$. 2

14. Write an algebraic expression for: a number y multiplied by 4 and then reduced by 9. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Evaluate $3p + 4q - 5$ for $p = 6$, $q = 2$. 3

16. Simplify $9x + 4 - 5x + 8 - 2$. 3

17. Find the sum of 23 and its reverse 32. Show that it is a multiple of 11. 3

OR

Evaluate $5p + 2q - 8$ for $p = 4$, $q = 7$. 3

18. Simplify $12x + 3 - 7x + 6 - 4$. 3

OR

Find the sum of 45 and its reverse 54. Show that it is a multiple of 11. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Think of a number n. Multiply it by 3, add 5 and then subtract 5. Show the result algebraically. What is the result when $n = 4$? 5

OR

In a number pyramid each block is the sum of the two blocks below it. The bottom row is 2, 5, 3. Find the middle row and the top block. 5

20. Think of a number n. Multiply it by 4, add 7 and then subtract 7. Show the result algebraically. What is the result when $n = 6$? 5

OR

In a number pyramid each block is the sum of the two blocks below it. The bottom row is 4, 1, 6. Find the middle row and the top block. 5

END OF QUESTION PAPER — Total: 40 Marks

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